

Coding with Kanak

Data Warehouse Architecture Components & Implementation



Unit 3

Data Warehousing: Unit-3

Data Warehouse Architecture Components and Implementation

3.1 Data warehouse Components

A data warehouse is an integrated system that consolidates data from operational systems and external sources, providing valuable insights for decision-making. It includes various components that work together to store, manage and access data efficiently.

The main components of a data warehouse include:

1. Operational Source Systems

- Provide raw data from internal systems (e.g., relational databases like Informix oracle) and external sources.

- Serve as the primary input for the data warehouse.

2. Load Manager

- Manages the ETL (Extract, Transform, Load) process for data extraction and transformation.

- Prepares data for entry into the warehouse, ensuring it meets the required format.

3. Warehouse Manager

- Oversees data storage, aggregation and analysis within the data warehouse.
- Handles tasks like data de-normalization, backup, collection and optimization for better performance.

4. Query Manager

- Handles user queries within the data warehouse.
- Supports querying, reporting and data retrieval, with functionality dependent on the available end-user tools.

5. Detailed Data

- Stores granular, raw data for complex analysis and reporting.
- Provides comprehensive insights and supports in-depth research.

6. Summarized Data

- Stores predefined aggregations of detailed data for faster queries and reports.
- Provides high-level insights to assist in decision-making.

7. Archive and Backup Data

- Ensures data integrity and disaster recovery through regular backups and archival storage.
- Archives older, less frequently accessed data for future use or compliance.

8. Metadata

- Metadata contains information about data structure, source and transformation processes.
- Supports the ETL process, warehouse management and querying by providing essential

context for data.

9. End-User Access Tools

- Serve as the interface for users to interact with the data warehouse.
- Include analysis, reporting and data mining tools, enabling users to access, query and

derive insights from the data.

3.2 Data Warehouse Architecture

• **Data layer:** Data is extracted from your sources and then transformed and loaded into the bottom tier using ETL tools. The bottom tier consists of your database server, data marts, and data lakes. Metadata is created in this tier – and data integration tools, like data virtualization, are used to seamlessly combine and aggregate data.

• **Semantics layer:** In the middle tier, online analytical processing (OLAP) and online transactional processing (OLTP) servers restructure the data for fast, complex queries and analytics.

• **Analytics layer:** The top tier is the front- end client layer. It holds the data warehouse access tools that let users interact with data, create dashboards and reports, monitor KPIs, mine and analyze data, build apps, and more. This tier often includes a workbench or sandbox area for data exploration and new data model development.

3.2 Operational and Physical, Extraction, Transforming and Loading

3.2.1 Operational and Physical

The operational and physical aspects refer to the overall management, scheduling, and the physical movement of data through the pipeline.

- Operational:

This refers to the ongoing, day-to-day management of the ETL process, including scheduling, monitoring, and handling any failures or errors that occur during execution.

It's about the workflow and ensuring the process runs correctly and on schedule.

The procedures and processes used to perform the extraction, which can be online (directly from the source) or offline (using other tools and routines)

Physical:

This refers to the actual movement of data from its source to the target system. It involves the physical infrastructure and software that perform the data extraction, transformation, and loading, whether it's on-premises hardware or cloud-based systems.

A comprehensive, low-level method of data retrieval that creates a complete copy of a device's memory or storage, including deleted files. It is more time-consuming and requires more effort than logical extraction.

3.2.2 Extraction, Transformation, and Loading (ETL)

Extraction, Transformation, and Loading (ETL) is a data integration process where data is first Extracted from various sources, Transformed into a usable format, and then Loaded into a central repository like a data warehouse.

This process is crucial for data analysis, reporting, and machine learning because it cleans and structures raw data into a consistent and reliable format for business intelligence.

- Extraction:

This is the initial step of collecting raw data from all available sources, such as databases, APIs, or files, regardless of their structure or format.

- Transformation:

The extracted data is then processed to clean and convert it into a consistent, usable format. This involves tasks like data mapping, cleansing, applying business rules, and

performing calculations to meet the specific requirements of the target system. This is often done in a temporary "staging area" before loading into the final data warehouse.

- Loading:

The final step is to load the transformed data into the target system, which is typically a data warehouse, data lake, or another storage system. This organized data is now ready for analysis and reporting by business intelligence tools and other applications.

3.3 Components of an Oracle Data Warehouse

Oracle Data Warehousing provides a robust and scalable solution for organizations to efficiently manage their data while ensuring high performance and security. It provides an enterprise-grade solution for managing large volumes of data with high performance, scalability, and security. By following best practices such as optimized data modeling, efficient ETL strategies, performance tuning, and security measures, businesses can maximize the value of their data warehouse investments.

Organizations should also consider leveraging Oracle's cloud-based offerings like Autonomous Data Warehouse to further simplify management and enhance scalability. With the right architecture and strategies, Oracle Data Warehousing can empower businesses to drive data-driven decision-making effectively.

Oracle Data Warehousing is built on the Oracle Database, providing a powerful platform for analytical processing, reporting, and business intelligence (BI) applications. It supports high-performance queries, parallel processing, and integration with cloud-based solutions like Oracle Autonomous Data Warehouse (ADW).

3.3.1 Key Components of Oracle Data Warehouse Architecture

1. Data Sources

- Structured data from transactional databases (OLTP systems)
- Unstructured data from logs, social media, IoT devices, and other external sources

2. ETL (Extract, Transform, Load) Layer

- Oracle Data Integrator (ODI) or other ETL tools extract, clean, and transform data
- Data quality and consistency checks
- Staging area for data preprocessing

3. Data Warehouse Storage

- Relational database storage optimized for read-heavy workloads
- Partitioning, indexing, and compression for efficient query execution

4. Data Marts

- Subject-specific subsets of the data warehouse (e.g., sales, finance, HR)
- Supports department-level analytics and reporting

5. Analytics and Reporting Layer

- Oracle Business Intelligence (BI) tools
- Oracle Analytics Cloud (OAC)
- Machine Learning models for predictive analytics

6. Security and Governance

- Role-based access control (RBAC)
- Data encryption and masking
- Compliance with regulations (GDPR, HIPAA, etc.)

3.3.2 Best Practices for Oracle Data Warehousing

1. Optimize Data Modeling

- Use star schema or snowflake schema for efficient querying
- Implement fact and dimension tables for structured analysis

2. Efficient Data Loading Strategies

- Use Oracle Data Integrator (ODI) for automated ETL processes
- Enable bulk loading instead of row-by-row inserts
- Implement partitioning to optimize large data loads

3. Improve Query Performance

- Utilize materialized views for faster query execution
- Implement indexing strategies (B-tree, bitmap indexes) to enhance retrieval speed
- Leverage Oracle Exadata for performance tuning

4. Leverage Parallel Processing

- Utilize Oracle Parallel Execution for large-scale data processing
- Distribute workload using Oracle RAC (Real Application Clusters)

5. Ensure Data Security and Compliance

- Implement database encryption to protect sensitive data
- Use audit logging to track changes and access
- Apply role-based security to limit access to critical data

6. Adopt Cloud-Based Solutions

- Consider Oracle Autonomous Data Warehouse (ADW) for automated tuning and scaling
- Use Oracle Cloud Infrastructure (OCI) for cost-effective and scalable storage

7. Regular Monitoring and Maintenance

- Conduct performance tuning with tools like AWR (Automatic Workload Repository) and

SQL Tuning Advisor

- Schedule regular data purging and archival to maintain efficiency

3.4 Data Transformation Functions

Data transformation functions convert raw data into a usable format by applying techniques like cleaning, normalizing, and aggregating. These functions are used in ETL/ELT processes to standardize data from different sources, making it consistent, accurate, and ready for analysis, reporting, and machine learning.

Common data transformation functions and techniques

Data Cleaning

:

- Purpose: To identify and correct errors, remove duplicates, and ensure consistency in the data.

Example: Removing redundant entries, standardizing date formats, or correcting typos.

Data Aggregation:

- Purpose: To summarize data by calculating totals, averages, or other metrics.

Example: Calculating the total sales for each product or finding the average customer age per region.

Data Normalization:

- Purpose: To adjust values from different scales to a common scale, which can help prevent certain variables from dominating analyses.

Example: Rescaling a dataset of salaries and ages to a common range (e.g., 0 to 1).

Data Derivation

:

- Purpose: To create new data fields from existing ones.

Example: Calculating a new "profit" column by subtracting "cost" from "revenue."

Data Enrichment

:

- Purpose: To add new data to the existing dataset from external sources.

Example: Adding geographical information to customer addresses based on IP addresses.

Data Standardization

:

- Purpose: To ensure uniformity across the dataset, making it reliable and easier to use.

Example: Ensuring all state abbreviations are consistent (e.g., CA instead of Cali or California).

Data Validation:

- Purpose: To ensure data meets specific criteria and quality rules.

Example: Checking if all email addresses follow the correct format before they are accepted into the database.

3.5 DBA Responsibilities

DBA responsibilities in data warehousing and mining include managing the data warehouse infrastructure, ensuring data quality through Extract, Transform, Load (ETL) processes, maintaining security and performance, and implementing backup and recovery plans. They also handle database design, user access, and troubleshooting to support business intelligence and strategic decision-making.

Key responsibilities

- ETL management:

Responsible for the correct extraction, transformation, and loading of data from various

sources into the data warehouse.

- Performance tuning and monitoring:

Continuously monitor and optimize database performance, transaction volumes, and response times.

- Security and compliance:

Implement and manage security measures, define access controls, monitor for security breaches, and ensure compliance with data policies.

- Backup and recovery:

Design, implement, and test backup, archiving, and disaster recovery strategies to prevent data loss.

- Database design and maintenance:

Design the data warehouse structure, manage database objects, and maintain the overall health and functionality of the database.

- Troubleshooting:

Diagnose and resolve errors and issues to ensure database availability and integrity.

- User support and documentation:

Provide support to users, create documentation for the database, and assist with data dictionary creation.

- Technical oversight:

Install and maintain database applications and upgrades, and recommend new technologies.

3.6 Capacity Planning

Capacity planning in data warehousing and mining is the strategic process of forecasting resource needs like storage, compute, and network bandwidth to ensure a data warehouse can handle current and future data loads without performance degradation. It involves analyzing historical and projected data volumes, user activity, and query patterns to provision adequate infrastructure, optimize costs, and align resources with business demands.

Key components and considerations.

Predict future data volume:

Data warehouses grow relentlessly, so capacity planning must forecast future storage requirements to avoid running out of space.

Analyze user and query patterns:

Understanding user groups, their query frequency, and the average size of their queries helps determine the necessary CPU, memory, and network capacity.

Monitor resource utilization:

Continuous monitoring of current resource usage (CPU, memory, disk, network) is crucial for identifying bottlenecks and making informed decisions about upgrades or scaling.

Optimize performance:

The goal is to ensure sufficient resources to maintain performance, avoid service interruptions, and tune the data load and queries for efficiency.

Manage costs:

A key objective is to provision the right amount of resources to meet demand without overspending on unnecessary capacity.

Consider cloud migration:

For organizations looking to reduce costs, capacity planning includes deciding which data to move to the cloud and how to manage that transition.

Process and steps

Gather requirements:

Collect business requirements, which should include projected data sizes, number of users, and expected query load.

Establish a baseline:

Analyze current resource usage to understand the existing load and performance characteristics.

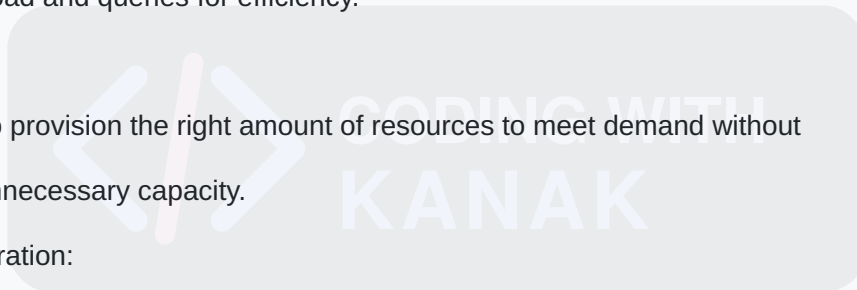
Forecast future needs:

Use historical data and projected growth to forecast future demand for storage, CPU, memory, and other resources.

Plan for infrastructure:

Determine the hardware, software, and other infrastructure components needed to meet future demand and ensure scalability.

Implement and monitor:



Provision the planned resources and continuously monitor performance to make adjustments as needed.

Tune and optimize:

Use the performance data to tune the data warehouse, optimize queries, and ensure the system is operating efficiently.

References:

[1]: <https://celerdata.com/glossary/data-warehouse>

[2]: <https://www.analytixlabs.co.in/blog/data-warehouse/>

[3]: <https://www.sap.com/products/data-cloud/datasphere/what-is-a-data-warehouse.html>

[4]: <https://www.ibm.com/think/topics/semantic-layer>

[5]: <https://www.thoughtspot.com/data-trends/data-modeling/conceptual-vs-logical-vs-physical-data-models>

[6]: <https://www.ibm.com/think/topics/etl>

[7]: https://docs.oracle.com/cd/B13789_01/server.101/b10736.pdf

[8]: <https://apps2fusion.com/old/oracle-fusion-online-training/fusion-middleware-development/oracle-data-integrator/863-introduction-to-oracle-data-warehouse-reference-architecture>

[9]: <https://www.bmc.com/blogs/dba-database-administrator/>

[10]: <https://www.tierpoint.com/blog/data-storage-capacity-planning/>

[11]: https://www.researchgate.net/publication/378148534_DATA_MINING_AND_DATA_WARE-

HOUSE_AN_IN-DEPTH_REVIEW